

MOOSA AHMED

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OBJECTIVE

Final-year Honours Computer Science student (Software Engineering & AI concentration) at Wilfrid Laurier University. Hands-on project experience in software development, database design, and applied AI/ML, grounded in coursework across Database Systems, Artificial Intelligence, and Software Engineering, with practical use of Python, SQL, and FastAPI.

SKILLS

Programming Languages: Python, Java, C, C#, JavaScript, MATLAB, R, SQL, HTML/CSS, ARM

Tools & Platforms: Git, VS Code, Eclipse, Jupyter, MySQL, PostgreSQL, Linux, Figma

Frameworks & Libraries: FastAPI, Entity Framework Core, Npgsql, Matplotlib, Tkinter

Business & Leadership: Team supervision, stakeholder communication, process improvement, Agile/Scrum, cross-functional collaboration

Concepts: Object-Oriented Programming, Data Structures & Algorithms, Reinforcement Learning, Database Design

EXPERIENCE

Backup Supervisor & Storefront Associate, Tim Hortons

Nov 2024 – Present

- Supervise team operations as Backup Supervisor, delegating tasks and ensuring staff adherence to company standards in a high-volume, fast-paced environment, directly applicable to managing priorities across rotational assignments.
- Handle cash transactions accurately while resolving customer concerns, maintaining food safety, and upholding cleanliness standards under time pressure.
- Deliver consistent, accurate service under pressure, helping the location maintain a positive guest experience during peak hours.

Peer Educator, Wilfrid Laurier University

Jan 2024 – May 2024

- Facilitated academic support sessions for international students, helping them adapt to Canadian post-secondary academic expectations.
- Mentored peers on campus resources, study strategies, and navigating college life effectively, building strong communication and coaching skills.
- Collaborated with faculty to develop and deliver educational materials, fostering an inclusive learning environment across a diverse cohort.

PROJECTS

Car Rental System

C#, PostgreSQL, Entity Framework, Npgsql

- Built a full-stack web application enabling customers to browse vehicles, make reservations, and manage bookings through a dynamic interface with a secure admin dashboard.
- Implemented back-end logic with C# and Entity Framework Core using Npgsql provider to connect to a PostgreSQL for vehicle availability tracking, pricing calculation, and role-based access control.

Smart Parking System

Python, Q-Learning, Dijkstra's, Matplotlib

- Modeled an urban resource-allocation problem as both a graph search task and a Markov Decision Process, implementing Dijkstra's algorithm with hill climbing and a Q-learning agent to compare deterministic optimization against adaptive decision-making.
- Designed and ran simulations across 3 constraint scenarios, evaluating cost, adaptability, and convergence; co-authored an 8-page IEEE-format research paper with a 4-person team and presented findings, experience translating technical analysis into a clear narrative for a non-technical audience.

Equipment Test Center DBMS

MySQL, Python, Tkinter

- Designed and implemented a relational database system for an equipment test center, using the MySQL Connector/ Python driver to interface with a normalized MySQL schema for data tracking and reporting.
- Built a Python/ Tkinter GUI as the front-end interface, working within a 2-person team to define entity relationship and deliver a functional end-to-end system, applying structured database design to a real-world operations use case

EDUCATION

Bachelor of Science in Computer Science, Wilfrid Laurier University

Expected Dec 2026

Concentration: Software Engineering & Artificial Intelligence

Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Engineering, Artificial Intelligence, Operating Systems, Cryptography, Computer Networks, UI/UX Design, Information Retrieval